Product Name: AMINOVEN 10% inf.	Territory: EXP	Colour: ● BLACK ● DIE CUT	1. Draft 05.06. 2018 10.20 Uhr		Variable Data:
Type of Packaging: PIL	Dosage: 500ml				
Material number: MOxxxxx/00 MY	2-D-Matrix Code x				
Pharma-Code (Laetus)	EAN Code:				
Dimension: 180 x 294 mm	Font: Antique Olive Smallest Size: 8 Pt.		Operator: Christian Nagy +43(0) 34 52 72266-22		

Aminoven 10%

Solution for infusion

QUALITATIVE AND QUANTITATIVE COMPOSITION
1000 ml contain:

Active constituents:

Isoleucine	5.00 g
Leucine	7.40 g
Lysine acetate	9.31 g
= Lysine	6.60 g
Methionine	4.30 g
Phenylalanine	5.10 g
Threonine	4.40 g
Tryptophan	2.00 g
Valine	6.20 g
Arginine	12.00 g
Histidine	3.00 g
Alanine	14.00 g
Glycine	11.00 g
Proline	11.20 g
Serine	6.50 g
Tyrosine	0.40 g
Taurine	1.00 g

Other constituents:
Glacial acetic acid
Water for injections

Total amino acids:	100.0 g/l
Total nitrogen:	16.2 g/l
Total energy:	1680 kJ/l (= 400 kcal/l)
pH-value:	5.5 - 6.3
Titration acidity:	22 mmol NaOH/l
Theoretical osmolarity:	990 mosm/l

Solution for intravenous infusion in glass bottle.
The solution is clear and colourless to slightly yellow.

Therapeutic indications
For supply of amino acids as part of a parenteral nutrition regimen.
Amino acid solutions should be administered generally in combination with adequate amount of energy supplements.

Contraindications
As for all amino acid solutions the administration of **Aminoven 10%** is contraindicated in the following conditions:
Disturbances of amino acid metabolism, metabolic acidosis, renal insufficiency without haemodialysis or haemofiltration treatment, advanced liver insufficiency, fluid overload, shock, hypoxia, decompensated heart failure.

The administration of **Aminoven 10%** is contraindicated in children less than 2 years of age.

Special warnings and precautions for use
Serum electrolytes, fluid balance and renal function should be monitored.

In cases of hypokalemia and/or hyponatremia adequate amounts of potassium and/or sodium should be supplied simultaneously.

Amino acid solutions may precipitate acute folate deficiency, folic acid should therefore be given daily.

Care should be exercised in the administration of large volume infusion fluids to patients with cardiac insufficiency.

No specific studies have been performed to assess the safety of **Aminoven 10%** in pregnancy or lactation. However, clinical experiences with similar parenteral amino acid solutions have shown no evidence of risk during pregnancy or breastfeeding. The risk/benefit relationship should be considered before administering **Aminoven 10%** during pregnancy or breast-feeding.

Infusion via peripheral veins in general can cause irritation of the vein wall and thrombophlebitis. Therefore, daily inspections of the insertion site are recommended.
If adjunction of lipid emulsions is indicated it should be administered when possible as a mixture with **Aminoven 10%** in order to minimise the risk of vein irritation.

The choice of a peripheral or central vein depends on the final osmolarity of the mixture. The general accepted limit for peripheral infusion is about 800 mosm/l, but it varies considerably with the age and the general condition of the patient and the characteristics of the peripheral veins.
Strict asepsis should be maintained, particularly when inserting a central vein catheter.

Aminoven 10% is for use as part of total parenteral nutrition regimen in combination with adequate amounts of energy supplements (carbohydrate solutions, fat emulsions), electrolytes, vitamins and trace elements.

Interactions with other medications
No interactions are known to date.

Due to the increased risk of microbiological contamination and incompatibilities, amino acid solutions should not be mixed with other drugs. When admixed with other nutrients like carbohydrates, lipid emulsions, electrolytes, vitamins and trace elements attention should be given to compatibility. Aseptic technique and thorough mixing should be used.

Compatibility data are available from the manufacturer for a number of mixtures.

Posology and method of administration

Posology
The daily requirement of amino acids depends on the body weight and the metabolic conditions of the patient.
The maximum daily dose varies with the clinical condition of the patient and may even change from day to day.
The recommended infusion period is to provide a continuous infusion for at least 14 hours up to 24 hours, depending on the clinical situation.
Bolus administration is not recommended.
The solution is administered as long as a parenteral nutrition is required.

Adults

Dosage:
10 - 20 ml of **Aminoven 10%** per kg body weight/ day (equivalent to 1.0 - 2.0 g amino acids per kg body weight/day), e.g. corresponding to 700 - 1400 ml **Aminoven 10%** at 70 kg body weight/day.

Maximum infusion rate:
1.0 ml of **Aminoven 10%** per kg body weight/ hour (equivalent to 0.1 g amino acids per kg body weight/ hour).

Maximum daily dose:
20 ml of **Aminoven 10%** per kg body weight/day (equivalent to 2.0 g amino acids per kg body weight/ day) corresponding to 1400 ml Aminoven 10% or 140 g amino acids at 70 kg body weight.

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Paediatric population
No studies have been performed in the paediatric population.
Aminoven 10% is contraindicated in children less than 2 years of age. For children under 2 years, paediatric amino acid preparations which are formulated to meet their different metabolic needs should be used.

Children and adolescents (2-18 years)

Dosage:
The dose should be adjusted to hydration status, biological development and body weight.

Maximum infusion rate:
Same as for adults, see information above.

Maximum daily dose:
Same as for adults, see information above.

Method of administration
For administration via a central vein as a continuous infusion.

Overdose (symptoms, emergency procedure, antidotes)
As with other amino acid solutions shivering, vomiting, nausea, and increased renal amino acid losses can occur when **Aminoven 10%** is given in overdose or the infusion rate is exceeded. Infusion should be stopped immediately in this case. It may be possible to continue with a reduced dosage.
A too rapid infusion can cause fluid overload and electrolyte disturbances.
There is no specific antidote for overdose. Emergency procedures should be general supportive measures, with particular attention to respiratory and cardiovascular systems. Close biochemical monitoring would be essential and specific abnormalities treated appropriately.

Undesirable effects
None known when correctly administered.
Those that occur during overdose (see above) are usually reversible and regress when therapy is discontinued. Infusion via peripheral veins in general can cause irritation of the vein wall and thrombophlebitis.
No other adverse events have been reported than these that can be seen in connection with parenteral nutrition in general.

Storage
Keep container in the outer carton. Do not store above 30°C. Do not freeze. Shelf life of the medicinal product as packaged for sale: 2 years.

Aminoven 10% should be used with sterile transfer equipment immediately after opening. Any unused solution should be discarded.
For single use only.

Do not use **Aminoven 10%** after expiry date.
Use only clear, particle-free solutions and undamaged containers.

Aminoven 10% may be aseptically admixed with other nutrients such as fat emulsions, carbohydrates and electrolytes. Chemical and physical stability data for a number of admixtures stored at 4°C for up to 9 days are available from the manufacturer upon request.

From a microbiological point of view, TPN admixtures compounded in uncontrolled or unvalidated conditions should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user and should normally be no longer than 24 hours at 2 to 8°C, unless mixing has taken place in controlled and validated aseptic conditions.

Pharmacological Properties:

Pharmacodynamics:
The amino acids contained in **Aminoven 10%** are all naturally occuring physiological compounds. As with the amino acids derived from the ingestion and assimilation of food proteins, parenterally administered amino acids enter the body pool of free amino acids and all subsequent metabolic pathways.

Pharmacokinetics:
The amino acids in **Aminoven 10%** enter the plasma pool of corresponding free amino acids. From the intravascular space, amino acids reach the interstitial fluid and the intracellular space of different tissues. Plasma and intracellular free amino acid concentrations are endogenously regulated for each single amino acid within narrow ranges, depending on the age, nutritional status and pathological condition of the patient. Balanced amino acid solutions such as **Aminoven 10%** do not significantly a lter t he physiologic a mino acid pool of essential and non-essential amino acids when infused at a constant and slow infusion rate. Characteristic changes in the physiologic plasma amino acid pool are only foreseeable when the regulative function of essential organs as liver and kidneys is seriously impaired. In such cases special formulated amino acid solutions may be recommended for restoring homeostasis.
Only a small proportion of the infused amino acids is eliminated by the kidneys.
For the majority of amino acids plasma half-lives between 10 and 30 minutes have been reported.

Packsizes
Bottles of 500 ml and 1000 ml

Jun 2018

Manufacturer
Fresenius Kabi Austria GmbH, Hafnerstrabe 36, 8055 Graz, Austria

Product Registration Holder/Importer
Fresenius Kabi Malaysia Sdn Bhd (535495-W)
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